

VOLUME 4

SOLUTION 0-2

Cicada 3301 - Liber Primus pages 0-2

Linked mirror geometry / IDEA OF THE END continuation



27 x 27 matrix / mirrored structures / Euler totient / Mobius phase / GP mod 29

SECTION 01

From NOW THE to IDEA

After NOW THE, the route ends at J(15,20). On the same diagonal there is the mirror A(8,13) - TH(11,16) - A(14,19), with radius 3.

Its center TH transforms through Euler's totient into U/V, giving the key structure A-U/V-A.

```
A-TH-A
TH -> phi(TH) = U/V
A-TH-A -> A-U/V-A
```

Mobius fixes $p=1$, so the active cyclic key is U/V-A-A. The next ciphertext runes on the same diagonal are:

```
J(15,20) -> E(16,19) -> D(17,18)
```

Using the fixed key:

```
J - U/V = I
E - A   = D
D - A   = EA
```

I-D-EA = IDEA

SECTION 02

IDEA endpoint and the J-D-J mirror

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27									
1	S/Z	H	E	O	G	M	IA/IO	F	1	S/Z	Y	E	NG	C/K	13	TH	J	G	AE	1	F	J	OE	1	O	N	TH	1	N	E	W	B				
2	A	S/Z	OE	EO	I	NG	U/V	1	N	AE	G	I	TH	1	H	B	N	IA/IO	P	B	N	1	EO	J	TH	1	A	NG	Y	X	1	B				
3	D	P	1	E	IA/IO	F	X	G	F	1	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	1	Y	H	1	B	U/V	1	G	O	P	1	NG			
4	AE	TH	1	I	AE	H	X	TH	P	G	1	Y	F	NG	C/K	J	1	TH	A	M	I	NG	I	IA/IO	Y	1	H	EO	1	T	H	AE				
5	IA/IO	1	X	Y	1	E	U/V	NG	C/K	F	N	1	EO	TH	J	I	TH	1	TH	P	1	Y	NG	EA	1	Y	N	E	Y	1	D	X	NG	W	1	
6	B	C/K	X	D	B	F	1	M	T	N	E	1	EA	J	N	L	G	B	1	X	G	TH	4	Y	I	D	A	1	NG	G	M	J				
7	R	O	L	EO	1	Y	T	P	I	TH	D	J	1	OE	H	L	1	EA	1	F	X	AE	P	G	X	1	F	L	W	EO	T	AE	1			
8	F	G	N	1	M	EO	L	N	1	NG	M	F	R	IA/IO	A	S/Z	1	EA	M	NG	X	X	EO	D	B	1	OE	D	M	O	1	EA				
9	EO	H	1	D	NG	G	1	OE	NG	L	U/V	R	N	T	1	AE	S/Z	Y	U/V	H	R	T	1	H	C/K	1	S/Z	OE	T	OE	NG	U/V				
10	R	1	S/Z	1	F	O	E	W	EA	OE	L	F	AE	1	M	R	NG	1	D	A	M	R	1	W	A	I	M	S/Z	N	1	S/Z	OE	R			
11	U/V	W	S/Z	L	IA/IO	4	OE	A	AE	NG	S/Z	D	P	T	1	C/K	R	TH	IA/IO	1	R	B	O	D	TH	F	1	Y	X	I	T	4				
12	OE	I	1	F	L	O	1	F	EA	1	R	O	OE	M	H	M	G	P	H	1	AE	H	N	O	H	Y	1	OE	S/Z	L	1	N	G	1		
13	AE	Y	1	OE	C/K	B	L	J	1	NG	L	OE	1	AE	J	EA	W	1	EA	TH	O	1	B	OE	Y	1	C/K	F	C/K	J	1	L	AE	4	N	1
14	TH	P	1	U/V	X	1	OE	X	G	1	P	F	S/Z	EO	AE	OE	1	NG	P	EO	O	E	A	1	G	AE	IA/IO	NG	TH	O	1	P	A			
15	OE	TH	R	NG	AE	1	C/K	S/Z	B	EO	Y	H	1	T	X	E	L	R	4	G	W	Y	1	J	F	I	N	IA/IO	C/K	Y	EA					
16	I	IA/IO	O	TH	1	E	C/K	AE	C/K	X	IA/IO	EA	1	O	L	C/K	1	F	R	D	NG	E	U/V	1	D	C/K	L	EA	S/Z	X	C/K					
17	G	IA/IO	W	S/Z	TH	1	F	D	NG	1	I	IA/IO	M	A	AE	G	1	W	S/Z	1	N	D	1	C/K	P	TH	X	P	EA	EA	1	W	U/V	1		
18	EA	W	1	F	W	J	Y	1	X	D	W	C/K	G	C/K	OE	1	D	X	OE	1	R	IA/IO	G	1	N	P	A	Y	P	1	C/K	Y	H	1		
19	F	E	J	EA	N	1	OE	AE	1	U/V	A	1	H	R	1	E	EA	TH	F	J	A	1	L	X	S/Z	T	1	M	F	L	F	T	1	G		
20	I	TH	1	U/V	L	G	1	X	EA	T	S/Z	L	J	P	1	L	X	I	1	OE	M	1	U/V	4	H	T	1	B	EO	L	D	H	B	M	1	
21	EA	R	B	1	N	H	B	E	G	EO	1	D	L	W	EO	IA/IO	P	O	1	H	E	EA	1	W	I	R	I	H	1	U/V	TH	H				
22	Y	1	N	X	B	G	J	P	U/V	1	NG	EA	F	N	I	E	D	IA/IO	NG	R	1	D	B	J	IA/IO	OE	M	I	1	F	T	1				
23	J	D	I	TH	R	L	S/Z	1	E	EO	O	G	B	T	D	1	TH	A	N	C/K	Y	1	IA/IO	S/Z	TH	D	1	NG	F	L	E					
24	G	H	C/K	1	E	O	I	T	N	X	1	P	T	F	H	R	D	E	F	T	1	AE	W	H	1	B	C/K	1	F	1	P	A	L			
25	U/V	EA	N	L	J	1	J	C/K	L	W	EA	D	U/V	D	EO	1	EA	X	D	W	H	EA	1	TH	IA/IO	AE	C/K	L	T	W						
26	E	I	C/K	1	P	OE	D	C/K	1	N	H	A	1	R	I	G	TH	EA	E	T	G	1	TH	H	O	IA/IO	W	AE	J	E	1	NG				
27	EA	D	1	O	AE	1	A	L	1	B	J	C/K	U/V	X	T	A	B	J	P	1	F	EA	1	H	D	N	IA/IO	U/V	P	S/Z	U/V	W	13			

The updated route after IDEA; D(17,18) becomes the center of the next mirror.

The final rune D(17,18) becomes the center of the next mirror:

J(15,20) -2- D(17,18) -2- J(19,16)

The previous endpoint J(15,20) is reused as one outer node. The opposite J(19,16) immediately becomes the center of another mirror.

OE(18,16) -1- J(19,16) -1- OE(20,16)

GEOMETRIC CONTINUITY

The route continues directly from one structure into the next, without jumping to an unrelated part of the grid.

SECTION 03

A direct bridge into OE-J-OE

The two mirrors are linked both geometrically and numerically. They share the node J(19,16), and the totient link is exact:

```
phi(D) = OE
J-D-J -> J-OE-J
```

The transformed J-D-J structure and the next OE-J-OE mirror then collapse to the same totient signature:

```
phi(J-OE-J) = 10-10-10
phi(OE-J-OE) = 10-10-10
```

EXACT MATCH

The next mirror not only continues directly from the previous one, but also preserves the same 10-10-10 totient signature.

This gives the transition two independent forms of continuity: one in the grid itself and one in the numerical signature.

SECTION 04

Decrypting OF THE

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27								
1	SZ	H	E	O	G	M	IAO	F	1	SZ	Y	E	NG	CK	13	TH	J	G	AE	1	F	J	OE	1	O	N	TH	1	N	E	W	B			
2	A	SZ	OE	EO	I	NG	UV	1	N	AE	G	I	TH	1	H	B	N	IAO	1	P	B	N	1	EO	J	TH	1	A	NG	Y	X	1	B		
3	D	P	1	E	IAO	F	X	G	F	1	SZ	P	T	F	P	UV	NG	Y	NG	X	IAO	1	Y	H	1	B	UV	1	G	O	P	1	NG		
4	AE	TH	1	I	AE	H	X	TH	P	G	1	Y	F	NG	CK	J	1	TH	A	M	I	NG	I	IAO	Y	1	H	EO	1	T	H	AE			
5	IAO	X	Y	1	E	UV	NG	1	CK	F	N	1	EO	TH	J	I	TH	1	TH	P	1	Y	NG	EA	1	Y	N	E	Y	1	D	X	NG	W	1
6	B	CK	X	D	B	F	1	M	T	N	E	1	EA	J	N	L	G	B	1	X	G	TH	4	Y	I	D	A	1	NG	G	M	J			
7	R	O	L	EO	1	Y	T	P	I	TH	D	J	OE	H	L	1	EA	1	F	X	AE	P	G	X	1	F	L	W	EO	T	AE	1			
8	F	G	N	1	M	EO	L	N	1	NG	M	F	R	IAO	1	A	SZ	1	EA	M	NG	X	X	EO	D	B	1	OE	D	M	O	1	EA		
9	EO	H	1	D	NG	G	1	OE	NG	L	UV	R	N	T	1	AE	SZ	Y	UV	H	R	T	1	H	CK	1	SZ	OE	T	OE	NG	UV			
10	R	1	SZ	1	F	O	E	W	EA	OE	L	F	AE	1	M	R	NG	1	D	A	M	R	1	W	1	A	I	M	SZ	N	1	SZ	OE	R	
11	UV	W	SZ	L	IAO	OE	A	AE	NG	SZ	D	P	T	1	CK	R	TH	IAO	1	R	B	O	D	TH	F	1	Y	X	I	T	4				
12	OE	I	1	F	L	O	1	F	EA	1	R	O	OE	M	H	M	G	P	H	1	AE	H	N	O	H	Y	1	OE	SZ	L	1	N	G	1	
13	AE	Y	1	OE	CK	B	L	J	1	NG	L	OE	AE	J	EA	W	1	EA	TH	O	1	B	OE	Y	1	CK	F	CK	J	1	L	AE	4	N	1
14	TH	P	1	UV	X	1	OE	X	G	1	P	F	SZ	EO	AE	OE	1	SG	P	EO	O	E	A	1	G	AE	IAO	NG	TH	O	1	P	A		
15	OE	TH	R	NG	AE	1	CK	SZ	B	EO	Y	H	1	T	X	E	L	R	4	G	W	Y	1	J	F	I	N	IAO	CK	Y	EA				
16	I	IAO	1	O	TH	1	E	CK	AE	CK	X	IAO	EA	1	O	L	CK	1	F	R	D	NG	E	UV	1	D	CK	L	EA	SZ	X	CK			
17	G	IAO	4	W	SZ	TH	1	F	D	NG	1	I	IAO	M	A	AE	G	1	W	SZ	1	N	D	1	CK	P	TH	X	P	EA	EA	1	W	UV	1
18	EA	W	1	F	W	J	Y	1	X	D	W	CK	G	CK	OE	1	D	X	OE	1	R	IAO	G	1	N	P	A	Y	P	1	CK	Y	H	1	
19	F	E	J	EA	N	1	OE	AE	1	UV	A	1	H	R	1	E	EA	TH	F	J	A	1	L	X	SZ	T	1	M	F	L	F	T	1	G	
20	I	TH	1	UV	L	G	1	X	EA	T	SZ	L	J	P	1	L	X	I	POE	M	1	UV	4	H	T	1	B	EO	L	D	H	B	M	1	
21	EA	R	B	1	N	H	B	E	G	EO	1	D	L	W	EO	IAO	P	O	1	H	E	EA	1	W	I	R	I	H	1	UV	TH	H			
22	Y	1	N	X	B	G	J	P	UV	1	NG	EA	F	N	I	E	D	IAO	NG	R	1	D	B	J	IAO	OE	M	I	1	F	T	1			
23	J	D	I	TH	R	L	SZ	1	E	EO	O	G	B	T	D	1	TH	A	N	CK	Y	1	IAO	SZ	TH	D	1	NG	F	L	E				
24	G	H	CK	1	E	O	I	T	N	X	1	P	T	F	H	R	D	E	F	T	1	AE	W	H	1	B	CK	1	F	1	P	A	L		
25	UV	EA	N	L	J	1	J	CK	L	W	EA	D	UV	D	EO	1	EA	X	D	W	H	EA	1	TH	IAO	AE	CK	L	T	W					
26	E	I	CK	1	P	OE	D	CK	1	N	H	A	1	R	I	G	TH	EA	E	T	G	1	TH	H	O	IAO	W	AE	J	E	1	NG			
27	EA	D	1	O	AE	1	A	L	1	B	J	CK	UV	X	T	A	B	J	P	1	F	EA	1	H	D	N	IAO	UV	P	SZ	UV	W	13		

Matrix state around the OE-J-OE mirror and the new plaintext branch.

The next words are linked through the OE(18,16) - J(19,16) - OE(20,16) mirror.

OF

First, OF is decrypted with the key produced by the preceding J-D-J structure: J-OE-J, with p=0.

$$\begin{aligned} X(18,15) - J &= 0 \\ OE(18,16) - OE &= F \end{aligned}$$

THE

The endpoint OE(18,16) is the outer node of the next OE-J-OE mirror. Applying Euler's totient to its center gives J -> I, producing OE-I-OE. Mobius fixes p=2, so the active key becomes OE-OE-I.

$$\begin{aligned} A(19,17) - OE &= TH \\ J(19,16) - OE &= E \end{aligned}$$

OF THE

SECTION 05

The route remains local

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27									
1	S/Z	H	E	O	G	M	IA/IO	F	1	S/Z	Y	E	NG	C/K	13	TH	J	G	AE	1	F	J	OE	1	O	N	TH	1	N	E	W	B				
2	A	S/Z	OE	EO	I	NG	U/V	1	N	AE	G	I	TH	1	H	B	N	IA/IO	1	P	B	N	1	EO	J	TH	1	A	NG	Y	X	1	B			
3	D	P	1	E	IA/IO	F	X	G	F	1	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	1	Y	H	1	B	U/V	1	G	O	P	1	NG			
4	AE	TH	1	I	AE	H	X	TH	P	G	1	Y	F	NG	C/K	J	1	TH	A	M	I	NG	I	IA/IO	Y	1	H	EO	1	T	H	AE				
5	IA/IO	X	Y	1	E	U/V	NG	1	C/K	F	N	1	EO	TH	J	I	TH	1	TH	P	1	Y	NG	EA	1	Y	N	E	Y	1	D	X	NG	W	1	
6	B	C/K	X	D	B	F	1	M	T	N	E	1	EA	J	N	L	G	B	1	X	G	TH	4	Y	I	D	A	1	NG	G	M	J				
7	R	O	L	EO	1	Y	T	P	I	TH	D	J	1	OE	H	L	1	EA	1	F	X	AE	P	G	X	1	F	L	W	EO	T	AE	1			
8	F	G	N	1	M	EO	L	N	1	NG	M	F	R	IA/IO	1	A	S/Z	1	EA	M	NG	X	X	EO	D	B	1	OE	D	M	O	1	EA			
9	EO	H	1	D	NG	G	1	OE	NG	L	U/V	R	N	T	1	AE	S/Z	Y	U/V	H	R	T	1	H	C/K	1	S/Z	OE	T	OE	NG	U/V				
10	R	1	S/Z	1	F	O	E	W	EA	OE	L	F	AE	1	M	R	NG	1	D	A	M	R	1	W	▶	A	I	M	S/Z	N	1	S/Z	OE	R		
11	U/V	W	S/Z	L	IA/IO	4	OE	A	AE	NG	S/Z	D	P	T	1	C/K	R	TH	IA/IO	1	R	B	O	D	TH	F	1	Y	X	I	T	4				
12	OE	I	1	F	L	O	1	F	EA	1	R	O	OE	M	H	M	G	P	H	1	AE	H	N	O	H	Y	1	OE	S/Z	L	1	N	G	1		
13	AE	Y	1	OE	C/K	B	L	J	1	NG	L	OE	1	AE	J	EA	W	1	EA	TH	O	1	B	OE	Y	1	C/K	F	C/K	J	1	L	AE	4	N	1
14	TH	P	1	U/V	X	1	OE	X	G	1	P	F	S/Z	EO	AE	OE	1	NG	P	EO	O	E	A	1	G	AE	IA/IO	NG	TH	O	1	P	A			
15	OE	TH	R	NG	AE	1	C/K	S/Z	B	EO	Y	H	1	T	X	E	L	R	4	G	W	Y	1	J	F	I	N	IA/IO	C/K	Y	EA					
16	I	IA/IO	1	O	TH	1	E	C/K	AE	C/K	X	IA/IO	EA	1	O	L	C/K	1	F	R	D	NG	E	U/V	1	D	C/K	L	EA	S/Z	X	C/K				
17	G	IA/IO	4	W	S/Z	TH	1	F	D	NG	1	I	IA/IO	M	A	AE	G	1	W	S/Z	1	N	D	1	C/K	P	TH	X	P	EA	EA	1	W	U/V	1	
18	EA	W	1	F	W	J	Y	1	X	D	W	C/K	G	C/K	OE	1	D	X	OE	1	R	IA/IO	G	1	N	P	A	Y	P	1	C/K	Y	H	1		
19	F	E	J	EA	N	1	OE	AE	1	U/V	A	1	H	R	1	E	EA	TH	F	J	A	1	L	X	S/Z	T	1	M	F	L	F	T	1	G		
20	I	TH	1	U/V	L	G	1	X	EA	T	S/Z	L	J	P	1	L	X	I	▶	OE	M	1	U/V	4	H	T	1	B	EO	L	D	H	B	M	1	
21	EA	R	B	1	N	H	B	E	G	EO	1	D	L	W	EO	IA/IO	P	O	1	H	E	EA	1	W	I	R	I	H	1	U/V	TH	H				
22	Y	1	N	X	B	G	J	P	U/V	1	NG	EA	F	N	I	E	D	IA/IO	NG	R	1	D	B	J	IA/IO	OE	M	I	1	F	T	1				
23	J	D	I	TH	R	L	S/Z	1	E	EO	O	G	B	T	D	1	TH	A	N	C/K	Y	1	IA/IO	S/Z	TH	D	1	NG	F	L	E					
24	G	H	C/K	1	E	O	I	T	N	X	1	P	T	F	H	R	D	E	F	T	1	AE	W	H	1	B	C/K	1	F	1	P	A	L			
25	U/V	EA	N	L	J	1	J	C/K	L	W	EA	D	U/V	D	EO	1	EA	X	D	W	H	EA	1	TH	IA/IO	AE	C/K	L	T	W						
26	E	I	C/K	1	P	OE	D	C/K	1	N	H	A	1	R	I	G	TH	EA	E	T	G	1	TH	H	O	IA/IO	W	AE	J	E	1	NG				
27	EA	D	1	O	AE	1	A	L	1	B	J	C/K	U/V	X	T	A	B	J	P	1	F	EA	1	H	D	N	IA/IO	U/V	P	S/Z	U/V	W	13			

Updated 27 x 27 matrix after the OF THE continuation.

ROUTE STATE

The newly recovered words remain attached to the same local cluster of linked mirrors rather than requiring a jump elsewhere in the matrix.

SECTION 06

Radius-4 geometry beyond THE

The rune F(19,15), immediately beyond THE, acts as a geometric pointer rather than continuing the word. It is the outer node of a radius-4 mirror:

F(19,15) -4- X(19,19) -4- F(19,23)

The same center X(19,19) also forms a perpendicular radius-4 mirror:

Y(15,19) -4- X(19,19) -4- Y(23,19)

At the same time, the endpoint J(19,16) belongs to another radius-4 structure:

J(19,16) -4- B(23,12) -4- J(27,8)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27						
1	SZ	H	E	O	G	M	IAR	F	SZ	Y	E	NG	CK	TH	J	G	AE	F	J	OE	O	N	TH	N	E	W	B						
2	A	SZ	OE	EO	I	NG	UV	N	AE	G	I	TH	H	B	N	IAR	P	B	N	EO	J	TH	A	NG	Y	X	T	B					
3	D	P	E	IAR	F	X	G	F	SZ	P	T	F	P	UV	NG	Y	NG	X	IAR	Y	H	B	UV	G	O	P	NG						
4	AE	TH	I	AE	H	X	TH	P	G	I	Y	F	NG	CK	J	TH	A	M	I	NG	I	IAR	Y	H	EO	I	T	H	AE				
5	IAR	X	Y	I	E	UV	NG	CK	F	N	EO	TH	J	I	TH	P	Y	NG	EA	Y	N	E	Y	D	X	NG	W	I					
6	B	CK	X	D	B	F	M	T	N	E	EA	J	N	L	G	B	X	G	TH	Y	I	D	A	NG	G	M	J						
7	R	O	L	EO	Y	T	P	I	TH	D	J	OE	H	L	EA	F	X	AE	P	G	X	F	L	W	EO	T	AE	I					
8	F	G	N	I	M	EO	L	N	NG	M	F	R	IAR	A	SZ	EA	M	NG	X	X	EO	D	B	OE	D	M	O	EA					
9	EO	H	I	D	NG	G	OE	NG	L	UV	R	N	T	AE	SZ	Y	UV	H	R	T	H	CK	SZ	OE	T	OE	NG	UV					
10	R	I	SZ	F	O	E	W	EA	OE	L	F	AE	M	R	NG	D	A	M	R	W	A	I	M	SZ	N	SZ	OE	R					
11	UV	W	SZ	L	IAR	OE	A	AE	NG	SZ	D	P	T	CK	R	TH	IAR	R	B	O	D	TH	F	Y	X	I	T	4					
12	OE	I	F	L	O	I	F	EA	R	O	OE	M	H	M	G	P	H	AE	H	N	O	H	Y	OE	SZ	L	N	G	I				
13	AE	Y	OE	CK	B	L	J	NG	L	OE	AE	J	EA	W	EA	TH	O	B	OE	Y	CK	F	CK	J	L	AE	N	I					
14	TH	P	I	UV	X	OE	X	G	I	P	F	SZ	EO	AE	OE	NG	P	EO	O	E	A	I	G	AE	IAR	NG	TH	O	P	A			
15	OE	TH	R	NG	AE	CK	SZ	B	EO	Y	H	T	X	E	L	R	4	G	W	Y	J	F	I	N	IAR	CK	Y	EA					
16	I	IAR	O	TH	E	CK	AE	CK	X	IAR	EA	O	L	CK	F	R	D	NG	E	UV	D	CK	L	EA	SZ	X	CK						
17	G	IAR	W	SZ	TH	I	F	D	NG	I	IAR	M	A	AE	G	I	W	SZ	N	D	CK	P	TH	X	P	EA	EA	I	W	UV			
18	EA	W	I	F	W	J	Y	X	D	W	CK	G	CK	OE	D	X	OE	I	R	IAR	G	I	N	P	A	Y	P	CK	Y	H	I		
19	F	E	J	EA	N	OE	AE	UV	A	H	R	I	E	EA	TH	F	J	A	L	X	SZ	T	I	M	F	L	F	T	I	G			
20	I	TH	I	UV	L	G	X	EA	T	SZ	L	J	P	I	L	X	I	OE	M	I	UV	4	H	T	B	EO	L	D	H	B	M	I	
21	EA	R	B	I	N	H	B	E	G	EO	D	L	W	EO	IAR	P	O	H	E	EA	I	W	I	R	I	H	I	UV	TH	H			
22	Y	I	N	X	B	G	J	P	UV	NG	EA	F	N	I	E	D	IAR	NG	R	I	D	B	J	IAR	OE	M	I	F	T	I			
23	J	D	I	TH	R	L	SZ	I	E	EO	O	G	B	T	D	I	TH	A	N	CK	Y	IAR	SZ	TH	D	NG	F	L	E				
24	G	H	CK	I	E	O	I	T	N	X	I	P	T	F	H	R	D	E	F	T	AE	W	H	I	B	CK	I	F	P	A	L		
25	UV	EA	N	L	J	J	CK	L	W	EA	D	UV	D	EO	I	EA	X	D	W	H	EA	I	TH	IAR	AE	CK	L	T	W				
26	E	I	CK	I	P	OE	D	CK	I	N	H	A	I	R	I	G	TH	EA	E	T	G	I	TH	H	O	IAR	W	AE	J	E	I	NG	
27	EA	D	I	O	AE	I	A	L	I	B	J	CK	UV	X	T	A	B	J	P	I	F	EA	H	D	N	IAR	UV	P	SZ	UV	W	I	3

The repeated radius-4 structures around the active route.

The geometry repeatedly emphasizes the same radius 4, and the first F(19,15) points across the mirror to the opposite F(19,23), where the next plaintext begins.

SECTION 07

Symmetry around X(19,19)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27									
1	S/Z	H	E	O	G	M	IA/IO	F	1	S/Z	Y	E	NG	C/K	13	TH	J	G	AE	1	F	J	OE	1	O	N	TH	1	N	E	W	B				
2	A	S/Z	OE	EO	I	NG	U/V	1	N	AE	G	I	TH	1	H	B	N	IA/IO	1	P	B	N	1	EO	J	TH	1	A	NG	Y	X	1	B			
3	D	P	1	E	IA/IO	F	X	G	F	1	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	1	Y	H	1	B	U/V	1	G	O	P	1	NG			
4	AE	TH	1	I	AE	H	X	TH	P	G	1	Y	F	NG	C/K	J	1	TH	A	M	I	NG	I	IA/IO	Y	1	H	EO	1	T	H	AE				
5	IA/IO	1	X	Y	1	E	U/V	NG	1	C/K	F	N	1	EO	TH	J	I	TH	1	TH	P	1	Y	NG	EA	1	Y	N	E	Y	1	D	X	NG	W	1
6	B	C/K	X	D	B	F	1	M	T	N	E	1	EA	J	N	L	G	B	1	X	G	TH	4	Y	I	D	A	1	NG	G	M	J				
7	R	O	L	EO	1	Y	T	P	I	TH	D	J	1	OE	H	L	1	EA	1	F	X	AE	P	G	X	1	F	L	W	EO	T	AE	1			
8	F	G	N	1	M	EO	L	N	1	NG	M	F	R	IA/IO	1	A	S/Z	1	EA	M	NG	X	X	EO	D	B	1	OE	D	M	O	1	EA			
9	EO	H	1	D	NG	G	1	OE	NG	L	U/V	R	N	T	1	AE	S/Z	Y	U/V	H	R	T	1	H	C/K	1	S/Z	OE	T	OE	NG	U/V				
10	R	1	S/Z	1	F	O	E	W	EA	OE	L	F	AE	1	M	R	NG	1	D	A	M	R	1	W	1	A	I	M	S/Z	N	1	S/Z	OE	R		
11	U/V	W	S/Z	L	IA/IO	4	OE	A	AE	NG	S/Z	D	P	T	1	C/K	R	TH	IA/IO	1	R	B	O	D	TH	F	1	Y	X	I	T	4				
12	OE	I	1	F	L	O	1	F	EA	1	R	O	OE	M	H	M	G	P	H	1	AE	H	N	O	H	Y	1	OE	S/Z	L	1	N	G	1		
13	AE	Y	1	OE	C/K	B	L	J	1	NG	L	OE	1	AE	J	EA	W	1	EA	TH	O	1	B	OE	Y	1	C/K	F	C/K	J	1	L	AE	4	N	1
14	TH	P	1	U/V	X	1	OE	X	G	1	P	F	S/Z	EO	AE	OE	1	NG	P	EO	O	E	A	1	G	AE	IA/IO	NG	TH	O	1	P	A			
15	OE	TH	R	NG	AE	1	C/K	S/Z	B	EO	Y	H	1	T	X	E	L	R	4	G	W	Y	1	J	F	I	N	IA/IO	C/K	Y	EA					
16	I	IA/IO	O	TH	1	E	C/K	AE	C/K	X	IA/IO	EA	1	O	L	C/K	1	F	R	D	NG	E	U/V	1	D	C/K	L	EA	S/Z	X	C/K					
17	G	IA/IO	W	S/Z	TH	1	F	D	NG	1	I	IA/IO	M	A	AE	G	1	W	S/Z	1	N	D	1	C/K	P	TH	X	P	EA	EA	1	W	U/V	1		
18	EA	W	1	F	W	J	Y	1	X	D	W	C/K	G	C/K	OE	1	D	X	OE	1	R	IA/IO	G	1	N	P	A	Y	P	1	C/K	Y	H	1		
19	F	E	J	EA	N	1	OE	AE	1	U/V	A	1	H	R	1	E	EA	TH	F	J	A	1	L	X	S/Z	T	1	M	F	L	F	T	1	G		
20	I	TH	1	U/V	L	G	1	X	EA	T	S/Z	L	J	P	1	L	X	I	1	OE	M	1	U/V	4	H	T	1	B	EO	L	D	H	B	M	1	
21	EA	R	B	1	N	H	B	E	G	EO	1	D	L	W	EO	IA/IO	P	O	1	H	E	EA	1	W	I	R	I	H	1	U/V	TH	H				
22	Y	1	N	X	B	G	J	P	U/V	1	NG	EA	F	N	I	E	D	IA/IO	NG	R	1	D	B	J	IA/IO	OE	M	I	1	F	T	1	1			
23	J	D	I	TH	R	L	S/Z	1	E	EO	O	G	B	T	D	1	TH	A	N	C/K	Y	1	IA/IO	S/Z	TH	D	1	NG	F	L	E					
24	G	H	C/K	1	E	O	I	T	N	X	1	P	T	F	H	R	D	E	F	T	1	AE	W	H	1	B	C/K	1	F	1	P	A	L			
25	U/V	EA	N	L	J	1	J	C/K	L	W	EA	D	U/V	D	EO	1	EA	X	D	W	H	EA	1	TH	IA/IO	AE	C/K	L	T	W						
26	E	I	C/K	1	P	OE	D	C/K	1	N	H	A	1	R	I	G	TH	EA	E	T	G	1	TH	H	O	IA/IO	W	AE	J	E	1	NG				
27	EA	D	1	O	AE	1	A	L	1	B	J	C/K	U/V	X	T	A	B	J	P	1	F	EA	1	H	D	N	IA/IO	U/V	P	S/Z	U/V	W	13			

The X(19,19) center and the surrounding radius-4 geometry.

X is at (19,19), and the symmetry around it is especially striking. The point (15,19) aligns cleanly with the surrounding purple cells.

STRUCTURAL FIT

This clean geometric alignment looks intentional and may be a strong indication that the route is following a real underlying structure.

The radius-4 mirror centered on X(19,19) connects F(19,15) with the opposite point F(19,23), so the geometry naturally carries the route across to the second F.

SECTION 08

Decrypting END

At the same time, J(19,16) belongs to another radius-4 mirror:

J(19,16) -4- B(23,12) -4- J(27,8)

Applying Euler's totient to the center gives B -> T, so the key structure becomes J-T-J. Mobius fixes $p=2$, giving the active key J-J-T.

J-B-J -> J-T-J
phase = 2
active key = J-J-T

From F(19,23), the next ciphertext runes are:

F(19,23) -> L(20,23) -> I(21,23)

Using the fixed key:

F - J = E
L - J = N
I - T = D

END

SECTION 09

END enters the next mirror

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27									
1	S/Z	H	E	O	G	M	IA/IO	F	1	S/Z	Y	E	NG	C/K	13	TH	J	G	AE	1	F	J	OE	1	O	N	TH	1	N	E	W	B				
2	A	S/Z	OE	EO	I	NG	U/V	1	N	AE	G	I	TH	1	H	B	N	IA/IO	P	B	N	1	EO	J	TH	1	A	NG	Y	X	1	B				
3	D	P	1	E	IA/IO	F	X	G	F	1	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	1	Y	H	1	B	U/V	1	G	O	P	1	NG			
4	AE	TH	1	I	AE	H	X	TH	P	G	1	Y	F	NG	C/K	J	1	TH	A	M	I	NG	I	IA/IO	Y	1	H	EO	1	T	H	AE				
5	IA/IO	X	Y	1	E	U/V	NG	1	C/K	F	N	1	EO	TH	J	I	TH	1	TH	P	1	Y	NG	EA	1	Y	N	E	Y	1	D	X	NG	W	1	
6	B	C/K	X	D	B	F	1	M	T	N	E	1	EA	J	N	L	G	B	1	X	G	TH	4	Y	I	D	A	1	NG	G	M	J				
7	R	O	L	EO	1	Y	T	P	I	TH	D	J	1	OE	H	L	1	EA	1	F	X	AE	P	G	X	1	F	L	W	EO	T	AE	1			
8	F	G	N	1	M	EO	L	N	1	NG	M	F	R	IA/IO	1	A	S/Z	1	EA	M	NG	X	X	EO	D	B	1	OE	D	M	O	1	EA			
9	EO	H	1	D	NG	G	1	OE	NG	L	U/V	R	N	T	1	AE	S/Z	Y	U/V	H	R	T	1	H	C/K	1	S/Z	OE	T	OE	NG	U/V				
10	R	1	S/Z	1	F	O	E	W	EA	OE	L	F	AE	1	M	R	NG	1	D	A	M	R	1	W	A	I	M	S/Z	N	1	S/Z	OE	R			
11	U/V	W	S/Z	L	IA/IO	1	OE	A	AE	NG	S/Z	D	P	T	1	C/K	R	TH	IA/IO	1	R	B	O	D	TH	F	1	Y	X	I	T	4				
12	OE	I	1	F	L	O	1	F	EA	1	R	O	OE	M	H	M	G	P	H	1	AE	H	N	O	H	Y	1	OE	S/Z	L	1	N	G	1		
13	AE	Y	1	OE	C/K	B	L	J	1	NG	L	OE	1	AE	J	EA	W	1	EA	TH	O	1	B	OE	Y	1	C/K	F	C/K	J	1	L	AE	4	N	1
14	TH	P	1	U/V	X	1	OE	X	G	1	P	F	S/Z	EO	AE	OE	1	NG	P	EO	O	E	A	1	G	AE	IA/IO	NG	TH	O	1	P	A			
15	OE	TH	R	NG	AE	1	C/K	S/Z	B	EO	Y	H	1	T	X	E	L	R	4	G	W	Y	1	J	F	I	N	IA/IO	C/K	Y	EA					
16	I	IA/IO	1	O	TH	1	E	C/K	AE	C/K	X	IA/IO	EA	1	O	L	C/K	1	F	R	D	NG	E	U/V	1	D	C/K	L	EA	S/Z	X	C/K				
17	G	IA/IO	1	W	S/Z	TH	1	F	D	NG	1	I	IA/IO	M	A	AE	G	1	W	S/Z	1	N	D	1	C/K	P	TH	X	P	EA	EA	1	W	U/V	1	
18	EA	W	1	F	W	J	Y	1	X	D	W	C/K	G	C/K	OE	1	D	X	OE	1	R	IA/IO	G	1	N	P	A	Y	P	1	C/K	Y	H	1		
19	F	E	J	EA	N	1	OE	AE	1	U/V	A	1	H	R	1	E	EA	TH	F	J	A	1	L	X	S/Z	T	1	M	F	L	F	T	1	G		
20	I	TH	1	U/V	L	G	1	X	EA	T	S/Z	L	J	P	1	L	X	I	1	OE	M	1	U/V	4	H	T	1	B	EO	L	D	H	B	M	1	
21	EA	R	B	1	N	H	B	E	G	EO	1	D	L	W	EO	IA/IO	P	O	1	H	E	EA	1	W	I	R	I	H	1	U/V	TH	H				
22	Y	1	N	X	B	G	J	P	U/V	1	NG	EA	F	N	I	E	D	IA/IO	NG	R	1	D	B	J	IA/IO	OE	M	I	1	F	T	1				
23	J	D	I	TH	R	L	S/Z	1	E	EO	O	G	B	T	D	1	TH	A	N	C/K	Y	1	IA/IO	S/Z	TH	D	1	NG	F	L	E					
24	G	H	C/K	1	E	O	I	T	N	X	1	P	T	F	H	R	D	E	F	T	1	AE	W	H	1	B	C/K	1	F	P	A	L				
25	U/V	EA	N	L	J	1	J	C/K	L	W	EA	D	U/V	D	EO	1	EA	X	D	W	H	EA	1	TH	IA/IO	AE	C/K	L	T	W						
26	E	I	C/K	1	P	OE	D	C/K	1	N	H	A	1	R	I	G	TH	EA	E	T	G	1	TH	H	O	IA/IO	W	AE	J	E	1	NG				
27	EA	D	1	O	AE	1	A	L	1	B	J	C/K	U/V	X	T	A	B	J	P	1	F	EA	1	H	D	N	IA/IO	U/V	P	S/Z	U/V	W	13			

The final I(21,23) of END enters a new mirrored structure.

The final I(21,23) of END enters the next mirror:

$$I(21,21) - R(21,22) - I(21,23)$$

Since R transforms to TH through Euler's totient, the mirror becomes:

$$R \rightarrow \phi(R) = TH$$

$$I-R-I \rightarrow I-TH-I$$

NEXT STRUCTURAL LINK

The endpoint of END is therefore not isolated: it lands directly inside the next mirror structure.

SECTION 10

The repeated 4-1-4 signature

The new I-TH-I structure has the totient signature:

$$\text{phi}(\text{I-TH-I}) = 4-1-4$$

This is exactly equal to the earlier structure used in the I MAY branch:

$$\text{phi}(\text{H-U/V-H}) = 4-1-4$$

REPEATED SIGNATURE

Both mirror structures produce the same 4-1-4 totient signature, linking the current endpoint back to an earlier successful branch of the route.

The current connected plaintext is:

**AS I GO, THE WEATHER TURNS COLD.
I MAY CRY NOW. THE IDEA OF THE END...**

Volume 4 therefore extends the route through IDEA, OF THE, and END while continuing the same local mirror, totient, Mobius-phase, and mod-29 framework.